Quantum Computing H.W. 1. 2022-14643 Sung-Soo Kim

1. The purity of the density matrix is given by the formula: $Tr(\rho^2)$

$$Tr(\rho^2) = Tr\left(\sum_j \sum_k p_j p_k |\psi_j\rangle\langle\psi_j|\psi_k\rangle\langle\psi_k|\right) = \sum_j p_j \sum_k p_k |\langle\psi_j|\psi_k\rangle|^2 \leq 1$$

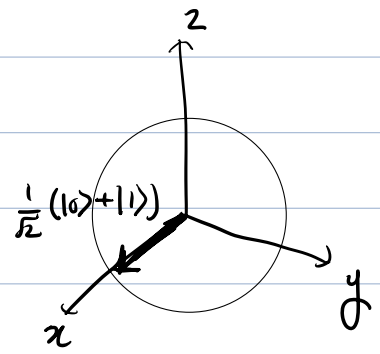
2. (a) $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

$$\langle\sigma_x\rangle = \langle\psi|\sigma_x|\psi\rangle = \frac{1}{2}\begin{pmatrix}1 & 1\end{pmatrix}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}1\\ 1\end{pmatrix} = 1$$

$$\langle\sigma_y\rangle = \langle\psi|\sigma_y|\psi\rangle = \frac{1}{2}\begin{pmatrix}1 & 1\end{pmatrix}\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}\begin{pmatrix}1\\ 1\end{pmatrix} = 0$$

$$\langle\sigma_z\rangle = \langle\psi|\sigma_z|\psi\rangle = \frac{1}{2}\begin{pmatrix}1 & 1\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}1\\ 1\end{pmatrix} = 0$$

Bloch vector:



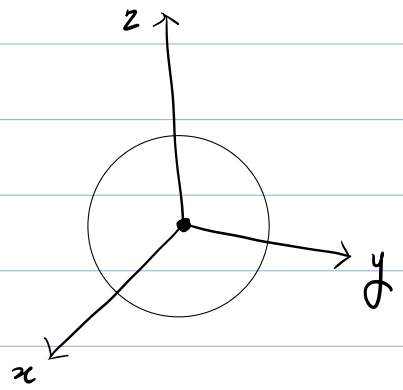
(b) $\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2}\mathbb{I}$.

$$\langle\sigma_x\rangle = Tr(\rho\sigma_x) = Tr\left(\frac{1}{2}\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\right) = \frac{1}{2}Tr\left(\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\right) = 0$$

$$\langle\sigma_y\rangle = Tr(\rho\sigma_y) = Tr\left(\frac{1}{2}\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}\right) = \frac{1}{2}Tr\left(\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}\right) = 0$$

$$\langle\sigma_z\rangle = Tr(\rho\sigma_z) = Tr\left(\frac{1}{2}\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\right) = \frac{1}{2}Tr\left(\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\right) = 0$$

Bloch vector:



3.

$$H_i = \frac{\mu^2}{r^3}\left(\sigma_{z.A}\otimes\sigma_{z.B} + \sigma_{y.A}\otimes\sigma_{y.B} - 2\,\sigma_{z.A}\otimes\sigma_{z.B}\right)$$

A. We write this in matrix form:

$$H_i = \frac{\mu^2}{r^3}\left(\begin{bmatrix}0&0&0&1\\0&0&1&0\\0&1&0&0\\1&0&0&0\end{bmatrix} + \begin{bmatrix}0&0&0&-1\\0&0&1&0\\0&1&0&0\\-1&0&0&0\end{bmatrix} - 2\begin{bmatrix}1&0&0&0\\0&-1&0&0\\0&0&-1&0\\0&0&0&1\end{bmatrix}\right)$$

$$= \frac{\mu^2}{r^3}\begin{bmatrix}-2&0&0&0\\0&2&2&0\\0&2&2&0\\0&0&0&-2\end{bmatrix}$$

$$(H-\lambda)|0\rangle = 0 \rightarrow \det(H-\lambda) = (-2-\lambda)(-2-\lambda)\left((2-\lambda)^2-4\right)$$

$$= (-2-\lambda)^2(-\lambda)(4-\lambda)$$

$$\therefore \lambda = 0,\ -2,\ 4$$

The allowed energies are $4\frac{\mu^2}{r^3}$, $-2\frac{\mu^2}{r^3}$, 0

$\lambda = 4 \quad v_1 = \frac{1}{\sqrt{2}}(0,1,1,0) \quad |\rho_1\rangle = v_1^T$

$\lambda = -2 \quad v_2 = (0,0,0,1) \quad |\rho_2\rangle = v_2^T$

$\lambda = -2 \quad v_3 = (1,0,0,0) \quad |\rho_3\rangle = v_3^T$

$\lambda = 0 \quad v_4 = \frac{1}{\sqrt{2}}(0,-1,1,0) \quad |\rho_4\rangle = v_4^T$

B.

$$|\rho_4\rangle\langle\rho_4| = v_4^T v_4 = \frac{1}{2}(0,-1,1,0)\begin{pmatrix}0\\-1\\1\\0\end{pmatrix} = \frac{1}{2}\begin{pmatrix}0&0&0&0\\0&1&-1&0\\0&-1&1&0\\0&0&0&0\end{pmatrix}$$

$$|\rho_3\rangle\langle\rho_3| = v_3^T v_3 = (1,0,0,0)\begin{pmatrix}1\\0\\0\\0\end{pmatrix} = \begin{pmatrix}1&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix}$$

$$|\rho_2\rangle\langle\rho_2| = v_2^T v_2 = (0,0,0,1)\begin{pmatrix}0\\0\\0\\1\end{pmatrix} = \begin{pmatrix}0&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&1\end{pmatrix}$$

$$|e_1\rangle\langle e_1| = v_1^T v_1 = \frac{1}{2}\begin{pmatrix} 0 & 1 & 1 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$H = \frac{\mu^2}{r^3}\left(4|e_1\rangle\langle e_1| - 2|e_2\rangle\langle e_2| - 2|e_3\rangle\langle e_3|\right)$$

4. $|GHZ\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle^{\otimes n} + |1\rangle^{\otimes n}\right)$, $|GHZ\rangle_3 = \frac{1}{\sqrt{2}}\left(|000\rangle + |111\rangle\right)$

$|0\rangle \otimes |0\rangle^{n-1} + |1\rangle \otimes |1\rangle$

A. $\rho_{000} = \frac{1}{\sqrt{2}}$ $\rho_{111} = \frac{1}{\sqrt{2}}$

Thus $E(\rho)$ can be computed as follows:

$E(\rho) = -\sum_{\mu} p_\mu \log p_\mu = -\left(\frac{1}{2}\log\left(\frac{1}{2}\right) + \frac{1}{2}\log\left(\frac{1}{2}\right)\right) = \log 2$

it is maximally entangled for 2-dimensional subsystems

B. $\rho = |GHZ\rangle\langle GHZ| = \frac{1}{2}\left(|000\rangle + |111\rangle\right)\left(\langle 000| + \langle 111|\right)$

$= \frac{1}{2}\left(|000\rangle\langle 000| + |111\rangle\langle 000| + |000\rangle\langle 111| + |111\rangle\langle 111|\right)$

$\langle 0_A|\rho|0_A\rangle = \frac{1}{2}|00_{BC}\rangle\langle 00_{BC}|$ $\langle 1_A|\rho|1_A\rangle = \frac{1}{2}|11_{BC}\rangle\langle 11_{BC}|$

The resultant state is then $\frac{1}{2}\left(|00\rangle\langle 00| + |11\rangle\langle 11|\right)$

$Tr_A(\rho^2) = \frac{1}{2} > \frac{1}{4}$ (dim 4)

$\therefore$ the system is partially mixed

C.

$\langle 0_B|\rho|0_B\rangle = \frac{1}{2}|00_{AC}\rangle\langle 00_{AC}|$, $\langle 1_B|\rho|1_B\rangle = \frac{1}{2}|11_{AC}\rangle\langle 11_{AC}|$

The resultant state is then $\frac{1}{2}\left(|00\rangle\langle 00| + |11\rangle\langle 11|\right)$

$Tr_B(\rho^2) = \frac{1}{2} > \frac{1}{4}$

$\therefore$ the system is partially mixed

$\langle 0_C|\rho|0_C\rangle = \frac{1}{2}|00_{AB}\rangle\langle 00_{AB}|$, $\langle 1_C|\rho|1_C\rangle = \frac{1}{2}|11_{AB}\rangle\langle 11_{AB}|$

The resultant state is then $\frac{1}{2}\left(|00\rangle\langle 00| + |11\rangle\langle 11|\right)$

$Tr_C(\rho^2) = \frac{1}{2} > \frac{1}{4}$

$\therefore$ the system is partially mixed

HW set 1 Problem #5

```qiskit
from qiskit.circuit.library import MCMTGate, XGate
from qiskit.circuit.library import ZGate, HGate
from qiskit.circuit.library import QFTGate, CZGate
from qiskit.circuit.library import CXGate, CCXGate
from qiskit.circuit.library import MCXGate
from qiskit.circuit.library import RZGate, RXGate
from qiskit.circuit.library import UGate
from math import pi, sqrt
from numpy.linalg import norm
from numpy.random import rand, randint
from numpy.random.mtrand import dirichlet
from numpy.fft import fft, ifft
from numpy.linalg.linalg import inv
from numpy.linalg.linalg import solve
from numpy.linalg.linalg import det
from numpy.linalg.linalg import eigvals
from numpy.linalg.linalg import eigvalsh
from numpy.linalg.linalg import eigvecs
from numpy.linalg.linalg import eigvecsh
from numpy.linalg.linalg import svd
from numpy.linalg.linalg import svdvals
from numpy.linalg.linalg import lstsq
from numpy.linalg.linalg import pinv
from numpy.linalg.linalg import matrix_rank
from numpy.linalg.linalg import cond
from numpy.linalg.linalg import norm
from numpy.linalg.linalg import slogdet
from numpy.linalg.linalg import trace
from numpy.linalg.linalg import multi_dot
from numpy.linalg.linalg import matrix_power
from numpy.linalg.linalg import kron
from numpy.linalg.linalg import tensordot
from numpy.linalg.linalg import einsum
from numpy.linalg.linalg import einsum_path
from numpy.linalg.linalg import inner
from numpy.linalg.linalg import outer
from numpy.linalg.linalg import vdot
from numpy.linalg.linalg import dot
from numpy.linalg.linalg import matmul
from numpy.linalg.linalg import cross
```

7.

(b) Theoretically, the quantum circuit above.

However. 20 qubits store a total of 2^{20} complex numbers.

This would require 2^{40} floats. and since $2^{40} \approx 10^{12}$, this exceeds 1 TB of memory: it is impossible on a personal computer.